

Finding Nicole's Ideal Home

Exploratory Data Analysis of King County Housing Data

Presented by: Sayed Kashif & Salah Haddad



The Agenda of our Presentation

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About Us

Salah Haddad

Product Manager

Sayed Kashif

Data Science Expert

Who is Nicole?

Our buyer persona and business question



Location

Central / lively neighborhood



Budget

Middle price range



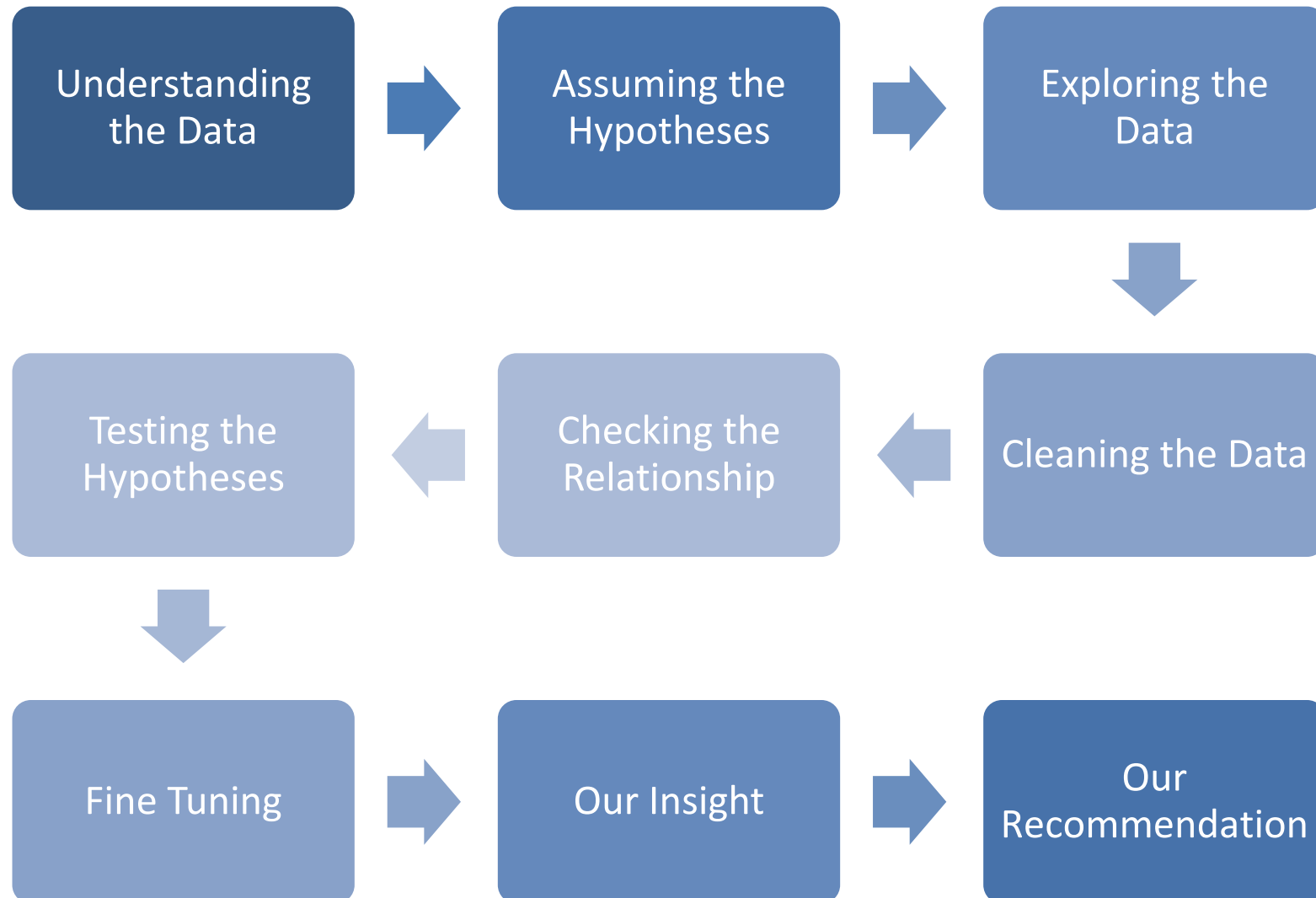
Timing

Plans to buy within 1 year

Business Question

What does the housing data tell us about the type of home Nicole should target?

Our Approach



Understanding the Data

Before testing the hypotheses, we inspect and clean the dataset

Dataset

The datasets were created by performing a left join between the **King County House Sales** and **King County House Details** tables.

Target

Mid rang price, lively central location in Seattle within a time one year

Main variables

price, bedrooms, bathrooms, sqft_living, zipcode

Quality checks

Missing values, duplicates, unusual values, outliers

Our 3 Hypotheses

Each hypothesis will be investigated using simple EDA

01 H1 — Price

Nicole prefers a home in the middle price range of the King County market.

02 H2 — Home characteristics

Homes in Nicole's price range offer a reasonable balance of living space, bedrooms and bathrooms.

03 H3 — Location

Some ZIP codes may offer a better balance between location and Nicole's middle-price budget.

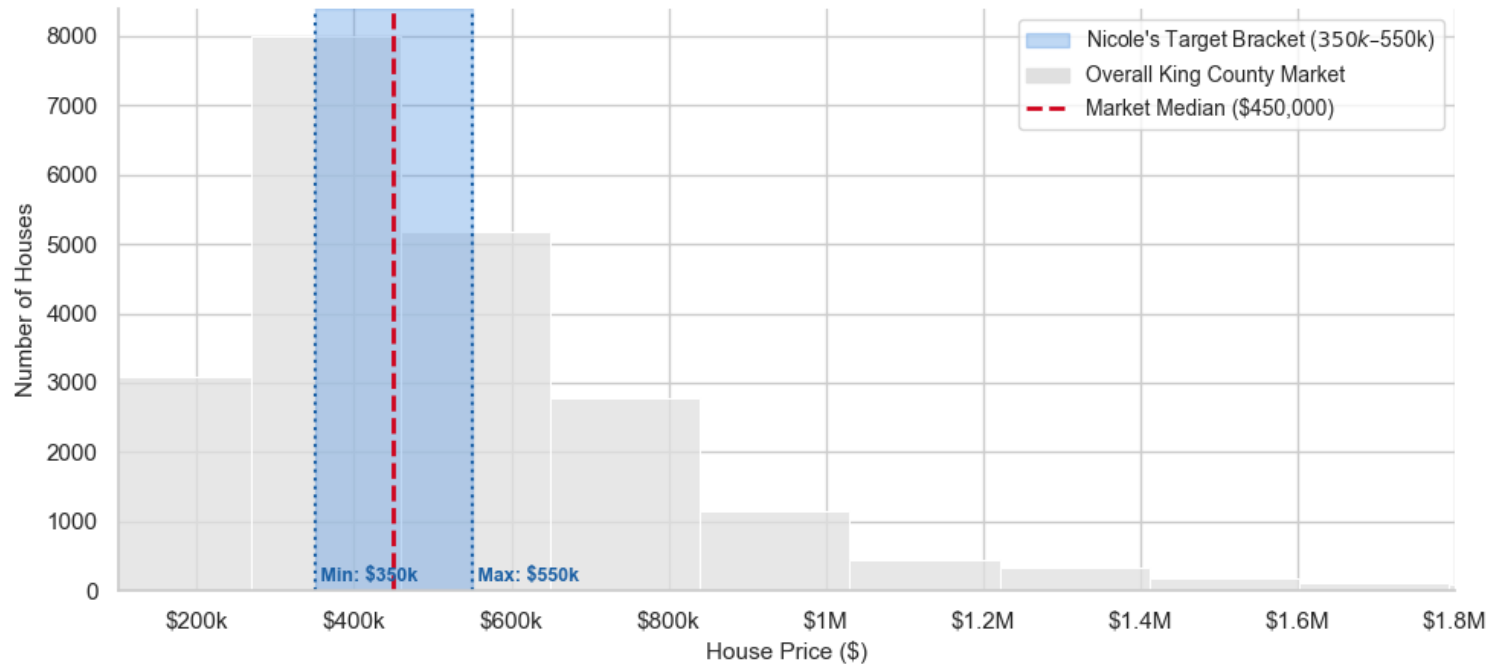
Limitation and Constraint

Due to presentation scope limitation, the analysis of time-based factors was excluded from this presentation.

H1 — What Does “Middle Price” Mean?

Use the distribution of price to define Nicole's target range

Hypothesis 1: King County Price Distribution & Mid-Market Target



Description

Nicole targets \$350k–\$550k, perfectly capturing the market's median price (\$450k) and its most concentrated area of house sales.

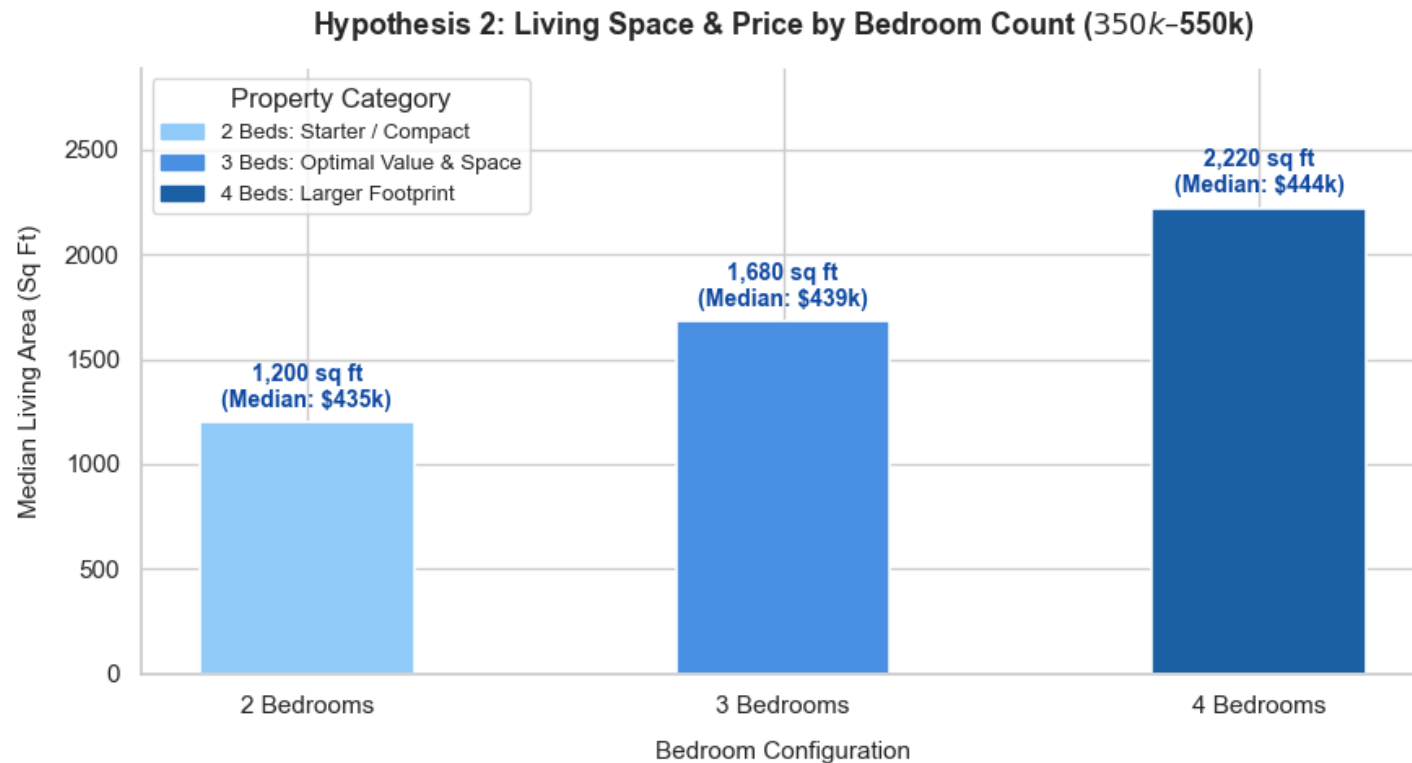
Our Finding

Calculate the actual **Min, Median and Max** price from the dataset.

Target range: \$ 350k – \$ 550k

H2 — Which sort of home Nicole gets for her Budget?

Explore the relationship between price and home characteristics



Description

The chart compares median living space and price, showing three-bedroom homes offer optimal value at 1,680 square feet for \$439k.

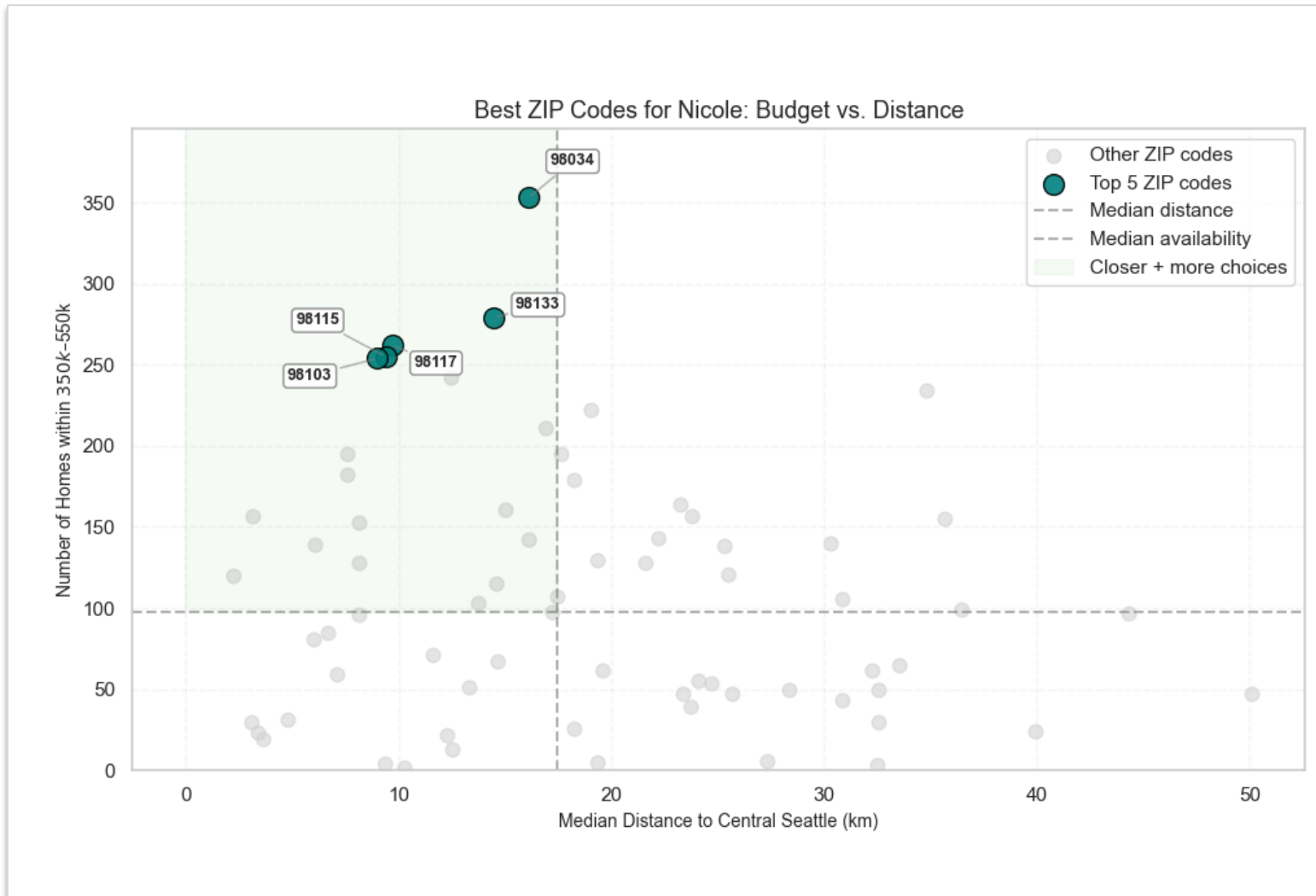
Our Finding

Use the **Vertical Bar Chart** to estimate the typical living area in Nicole's price range.

Target size: **1,680** sqft

H3 — Where Should Nicole Look?

Compare the number of houses within the Nicole's budget in association to Median distance to Central Seattle.



Description

The Scatter Plot compares home availability against commute distance, highlighting five top ZIP codes inside the optimal green quadrant.

Our Finding

We Identified the Five top ZIP codes (the optimal are: **98115 & 98103**) offering the highest housing availability closest to central Seattle within budget.

NOTE: The dataset does not directly measure "liveliness". ZIP code is only a location proxy.

Our Insights

 **Price**

\$350k – \$550k

 **Living space**

1,680 sqft

 **Bedrooms**

Optimal 3

 **Bathrooms**

NA

 **Location**

ZIP 98115 / 98103

 **Timing**

TBD

Our Recommendations

1st recommendation

Focus her search on homes priced between \$350K and \$550K

2nd recommendation

Nicole should target 3-bedroom homes around 1,680 sqft within her \$350K–\$550K budget.

3rd recommendation

Nicole should prioritize 98115 and 98103 when starting her home search.

Thank you for your Valuable Attention !



Q & A